

INCIDENT REPORT

Home SOC Lab — Attack Simulation & Detection Analysis

Prepared by: Lance Vidaure | Date: May 16, 2026 | Classification: LAB — UNCLASSIFIED

1. EXECUTIVE SUMMARY

On May 16, 2026, a controlled attack simulation was conducted within a home SOC lab environment consisting of three virtual machines: a pfSense firewall (192.168.10.1), a Windows Server 2022 domain controller (192.168.10.10), and an Ubuntu 22.04 server running Wazuh SIEM (192.168.10.20). A Kali Linux attacker VM (192.168.10.30) was deployed to simulate adversary behavior across three attack phases.

The simulation revealed that the Wazuh SIEM successfully detected brute-force SSH attacks and privilege escalation events, but failed to generate alerts for initial nmap reconnaissance scans — a detection gap consistent with real-world SIEM limitations. This report documents the full attack timeline, MITRE ATT&CK mappings, IOC inventory, detection findings, and remediation recommendations.

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Attacks Simulated	Detected by SIEM	Detection Gap Found	Rules Fired

2. INCIDENT DETAILS

Incident ID:	IR-2026-001
Date:	May 16, 2026
Time Range:	15:40 UTC — 19:57 UTC
Environment:	Home SOC Lab (VirtualBox — Internal Network: SOC-Lab-Network)
Analyst:	Lance Vidaure
Severity:	MEDIUM — Controlled simulation; no production systems affected
Status:	CLOSED — All attacks completed; findings documented

3. LAB ENVIRONMENT

Host	IP Address	Role	OS
pfSense-Firewall	192.168.10.1	Network Gateway / Firewall	pfSense CE 2.7.2
WinServer-DC	192.168.10.10	Domain Controller (SOCLAB)	Windows Server 2022
Ubuntu-Target	192.168.10.20	SIEM / Wazuh Manager + Indexer	Ubuntu 22.04 LTS

Kali-Attacker	192.168.10.30	Attacker / Red Team VM	Kali Linux 2026.1
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4. FIVE WS

WHO:	Attacker: Kali Linux VM (192.168.10.30) — simulated adversary. Target: Ubuntu-Target SIEM (192.168.10.20) and WinServer-DC (192.168.10.10).
WHAT:	Three-phase attack simulation: network reconnaissance (nmap), SSH brute-force (Hydra + Metasploit), and privilege escalation (new admin account creation on Windows Server).
WHEN:	May 16, 2026 — Attack Phase 1 at 15:40 UTC, Phase 2 at 15:40–15:41 UTC, Phase 3 at 16:51–16:59 UTC. Wazuh agent connected at 09:20 UTC.
WHERE:	Internal lab network SOC-Lab-Network (192.168.10.0/24) running inside VirtualBox on a physical Windows 11 host.
WHY:	Controlled simulation to test SIEM detection capabilities, identify detection gaps, and document findings for portfolio and GitHub documentation.

5. ATTACK TIMELINE

Time (UTC)	Event	Tool	Target
09:04	Wazuh agent installed on WinServer-DC	Manual	192.168.10.20:1515
09:20	Agent enrolled — valid key received	Wazuh	192.168.10.20
09:20	Agent connected to manager	Wazuh	192.168.10.20:1514
15:40	nmap service version scan	nmap -sV	192.168.10.20
15:40	SSH (22) and HTTPS (443) discovered	nmap	192.168.10.20
15:40	Hydra SSH brute-force (6 attempts)	Hydra	192.168.10.20:22
15:41	Hydra completed — 0 valid credentials	Hydra	192.168.10.20:22
15:41	Wazuh Rule 5760 fired — auth failed	Wazuh	192.168.10.20
15:41	Metasploit SSH version scan	MSF auxiliary/scanner/ssh/ssh_version	192.168.10.20
15:41	SSH fingerprint: OpenSSH 8.9p1 Ubuntu	Metasploit	192.168.10.20:22
15:56	Metasploit SSH brute-force (6 attempts)	MSF ssh_login	192.168.10.20:22
15:57	Wazuh Rules 5503 + 5760 fired	Wazuh	192.168.10.20
16:51	testadmin account created on WinServer-DC	net user	192.168.10.10
16:59	testadmin added to Administrators group	net localgroup	192.168.10.10
16:59	Wazuh Event IDs 4738 + 4732 captured	Wazuh	192.168.10.10

6. ATTACK DETAILS & DETECTION FINDINGS

Attack 1: Network Reconnaissance — nmap Service Version Scan		MEDIUM	MISSED
Tool:	nmap 7.x		
Command:	nmap -sV 192.168.10.20		
Source IP:	192.168.10.30 (Kali-Attacker)		
Target IP:	192.168.10.20 (Ubuntu-Target)		
Ports Found:	22/tcp — SSH (OpenSSH), 443/tcp — HTTPS (Wazuh Dashboard)		
MITRE ATT&CK::	T1046 — Network Service Discovery		
Wazuh Alert:	NONE — No alert generated		
Detection Gap:	Wazuh does not detect nmap scans by default. No port scan detection rule was configured.		

Attack 2: SSH Brute-Force — Hydra Password Attack		HIGH	DETECTED
Tool:	Hydra v9.6		
Command:	hydra -l root -P /tmp/passlist.txt ssh://192.168.10.20 -t 4 -V		
Source IP:	192.168.10.30 (Kali-Attacker)		
Target IP:	192.168.10.20:22 (Ubuntu-Target SSH)		
Attempts:	6 password attempts — admin, root, kali, password, Password123, wrongpass		
Result:	0 valid credentials found — SSH login blocked		
MITRE ATT&CK::	T1110 — Brute Force / T1110.001 — Password Guessing		
Wazuh Rules:	Rule 5760 (level 5) — sshd: authentication failed Rule 5503 (level 5) — PAM: User login failed		
Source Port:	59654, 59662 (multiple connections)		

Attack 3: Metasploit SSH Fingerprint + Brute-Force		HIGH	DETECTED
Tool:	Metasploit Framework — auxiliary/scanner/ssh/ssh_version + ssh_login		
Source IP:	192.168.10.30 (Kali-Attacker)		
Target IP:	192.168.10.20:22 (Ubuntu-Target SSH)		
SSH Version Found:	OpenSSH 8.9p1 Ubuntu-3ubuntu0.15 — Key: ssh-ed25519		
Encryption Found:	chacha20-poly1305, aes128-ctr, aes192-ctr, aes256-ctr, aes128-gcm, aes256-gcm		
Brute-Force:	6 attempts — root:admin, root:root, root:kali, root:password, root:Password123, root:wrongpass		
MITRE ATT&CK::	T1046 — Network Service Discovery T1592 — Gather Victim Host Information T1110 — Brute Force		
Wazuh Rules:	Rule 5503 (level 5) — PAM: User login failed Rule 5760 (level 5) — sshd: authentication failed		
Source Ports:	42003, 37743		

Attack 4: Privilege Escalation — New Admin Account Creation		CRITICAL	DETECTED
Action:	Created local user testadmin and added to Administrators group on WinServer-DC		
Commands:	net user testadmin Password123! /add net localgroup administrators testadmin /add		
Source:	Administrator account on WinServer-DC (192.168.10.10)		
MITRE ATT&CK::	T1136.001 — Create Account: Local Account T1078 — Valid Accounts		
Windows Event IDs:	4738 — User account changed 4732 — Member added to security-enabled local group		
Wazuh Detection:	Both events captured in alerts log with full account details, SID, and timestamps		
Target Username:	testadmin Target Group: Administrators (S-1-5-32-544)		

7. INDICATORS OF COMPROMISE (IOCS)

IOC Type	Value	Context
IP Address	192.168.10.30	Attacker — Kali Linux VM
IP Address	192.168.10.20	Target — Ubuntu SIEM
IP Address	192.168.10.10	Target — Windows Server DC
Port	22/tcp	SSH — targeted by Hydra and Metasploit
Port	443/tcp	HTTPS — discovered via nmap scan
Port	1515/tcp	Wazuh enrollment service
Port	1514/tcp	Wazuh agent communication
Username	root	Brute-force target on Ubuntu SSH
Username	testadmin	Newly created privileged account on WinServer-DC

Event ID	4738	User account changed — WinServer-DC
Event ID	4732	Member added to privileged group — WinServer-DC
Wazuh Rule	5760 (level 5)	sshd: authentication failed
Wazuh Rule	5503 (level 5)	PAM: User login failed
Wazuh Rule	60107 (level 4)	Failed attempt to perform privileged operation
Tool	nmap -sV	T1046 — Network Service Discovery
Tool	Hydra v9.6	T1110 — Brute Force
Tool	Metasploit ssh_version	T1592 — Gather Victim Host Information
Tool	Metasploit ssh_login	T1110 — Brute Force

8. MITRE ATT&CK; MAPPING

Technique ID	Technique Name	Tool Used	Detected
T1046	Network Service Discovery	nmap -sV	NO
T1592	Gather Victim Host Information	Metasploit ssh_version	NO
T1110	Brute Force	Hydra + Metasploit ssh_login	YES
T1110.001	Password Guessing	Hydra wordlist attack	YES
T1136.001	Create Account: Local Account	net user /add	YES
T1078	Valid Accounts	net localgroup administrators	YES

9. DETECTION GAPS & ANALYSIS

GAP IDENTIFIED: nmap Reconnaissance Not Detected	
Finding:	Kali ran nmap -sV against 192.168.10.20 at 15:40 UTC. Wazuh generated zero alerts for this scan.
Root Cause:	Wazuh does not include a default port scan detection rule. The scan occurred at the network level and was not logged by the Wazuh agent on Ubuntu.
Impact:	An attacker could fully enumerate all open services on monitored hosts without triggering any SIEM alert. This represents a blind spot in the detection architecture.
Real-World Context:	This gap is common in SIEM deployments that rely solely on host-based agents. Network-layer visibility (IDS/IPS, firewall logs) is required to detect reconnaissance activity.

10. REMEDIATION RECOMMENDATIONS

Add nmap/Port Scan Detection Rule	HIGH
Add a custom Wazuh rule to detect rapid connection attempts to multiple ports from a single source IP. Alternatively, integrate pfSense firewall logs into Wazuh to capture network-level reconnaissance activity.	
Increase Brute-Force Alert Level	MEDIUM
Custom rules 100002 (5+ failed logons in 2 minutes) were added but not triggered during this simulation because attempts were spread across time. Reduce the threshold to 3 attempts in 60 seconds to catch slower attacks.	
Enable Account Creation Alerting	HIGH
Wazuh detected Event IDs 4738 and 4732 but did not fire custom rule 100003. Verify the rule SID (60144) matches the actual Windows event rule in Wazuh. Consider adding explicit rules for Event IDs 4720 and 4732.	

Deploy Network IDS (Suricata)	HIGH
Install Suricata on pfSense or Ubuntu to provide network-layer threat detection. This closes the reconnaissance detection gap entirely and provides visibility into traffic Wazuh agents cannot see.	
Implement SSH Fail2Ban	MEDIUM
Install and configure Fail2Ban on Ubuntu to automatically block IPs after repeated SSH failures. This provides active response capability beyond passive SIEM alerting.	
Restrict Local Admin Creation	CRITICAL
Implement a GPO on WinServer-DC to prevent standard users from creating local administrator accounts. Alert on any new account creation via Wazuh custom rule targeting Event ID 4720.	

11. LESSONS LEARNED

1. Host-based SIEM agents alone are insufficient — network-layer visibility (IDS/IPS, firewall log ingestion) is required to detect reconnaissance.
2. Wazuh successfully detected brute-force attacks in real time, demonstrating the value of authentication log monitoring.
3. Privilege escalation events (Windows Event IDs 4720, 4732, 4738) are reliably captured by Wazuh when agents are properly enrolled.
4. Metasploit service fingerprinting (T1592) produced no alert — this is a gap that requires either network IDS or enhanced SSH logging (verbose mode).
5. RAM constraints on an 8GB host required VM rotation to run attack simulations — in a production SOC environment, dedicated hardware or cloud infrastructure eliminates this limitation.
6. Detection gaps identified in lab environments translate directly to real-world SIEM tuning priorities and are valuable portfolio documentation.

12. CONCLUSION

This simulation successfully validated the core detection capabilities of a home SOC lab built on Wazuh SIEM. The lab demonstrated real-time detection of SSH brute-force attacks and Windows privilege escalation events, while also identifying a meaningful detection gap in reconnaissance coverage. The findings directly inform the next phase of the lab build: deploying Suricata as a network IDS and integrating AWS CloudTrail for cloud security monitoring.

All attack tools, detection rules, IOCs, and MITRE ATT&CK mappings documented in this report are available in the SOC Lab GitHub repository at github.com/lanceanthony.